

Failure Mode & Root-Cause Analysis Report

Automated Industrial PPE Safety Monitoring System | Version 2.0 Evaluation

Model	YOLOv8 Nano (yolov8n.pt)
Test Evaluation	127 Site Images / 232 Worker Instances
Validation mAP@50	83.3% (overall, clean validation split)
System Latency	2.0 ms Inference / ~4.5 ms Total per Image

Executive Technical Clarification: Perception vs. Decision Engine

The underlying YOLOv8 object detection model operates with high accuracy (83.3% mAP@50) across physical classes. The false-violation edge cases identified below are, in most cases, not machine learning classification failures; rather, they represent boundary limitations of static post-processing spatial rules (e.g., assuming a worker's head always occupies the top 25% of their bounding box), input data quality issues, or known out-of-distribution behavior. Six distinct failure categories were identified through structured and random-sample testing, root-caused, and addressed where feasible. This version supersedes v1.0 with a corrected root-cause finding for EXIF rotation (Section 2) and two newly identified categories (Sections 4 and 6).

Root-Cause Breakdown of Identified Edge Cases

1. Non-Upright Posture & Bending Over (Spatial Rule Break)

STATUS: DOCUMENTED — Fix Planned (v2)

OBSERVED BEHAVIOR	Worker bent at approximately 90° while wearing full PPE is marked VIOLATION (Missing: Helmet), despite correct detection of all items.
MODEL PERCEPTION CHECK	YOLO correctly detected both the Person box and the Helmet box in the frame — confirmed NOT a detection failure.
ROOT-CAUSE MECHANICS	The engine calculates head location as $y1 + height * 0.25$, assuming standing posture. Crouching shifts the actual head position outside this fixed proportional region.
Production Fix	Upgrade from bounding-box heuristics to YOLOv8-Pose Estimation, tracking ear/nose keypoints so the head region maps to actual joint coordinates regardless of torso angle. Not yet implemented — requires integrating a separate pose model.

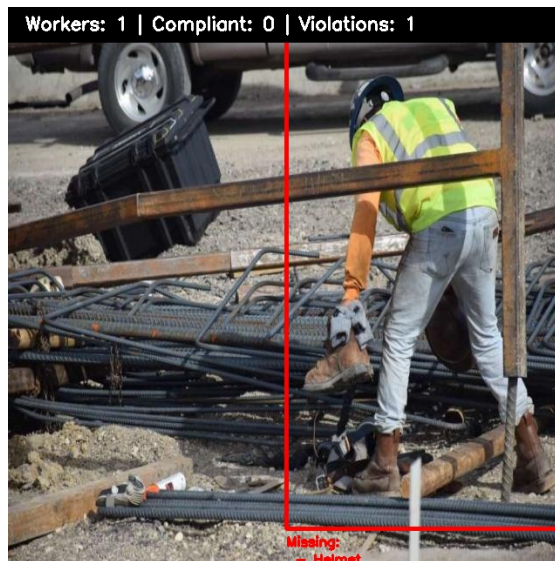


Fig. 1.1 — image810.jpg: crouching worker flagged as false helmet violation.

2. Rotated Image Feeds (Data Preprocessing Issue)

STATUS: PARTIALLY ADDRESSED — Root Cause Refined

OBSERVED BEHAVIOR	Images displaying workers in horizontal/rotated orientation caused the head-region heuristic to sample an incorrect area (e.g. shoulder instead of head), producing false VIOLATION results.
MODEL PERCEPTION CHECK	YOLO detects Person and PPE items correctly across the rotated coordinate axis — a perception/geometry mismatch, not a detection failure.
ROOT-CAUSE MECHANICS	Initial hypothesis: unread EXIF orientation metadata (raw OpenCV imread ignoring rotation tags). Verified finding: diagnostic testing via <code>PIL.Image.getexif()</code> confirmed affected images in this dataset carry NO embedded EXIF orientation metadata at all (empty EXIF dictionary). This is a materially different root cause: the images are not merely "unrotated despite having metadata" — they have no recoverable orientation data to act on, consistent with metadata being stripped during dataset pre-processing/re-export.
Production Fix	Standard EXIF-based auto-rotation (<code>PIL.ImageOps.exif_transpose</code>) was implemented and works correctly for any image carrying valid EXIF data. For metadata-free images, this fix has no effect (confirmed via testing). A general-purpose solution requires a visual, content-based orientation-detection model — scoped as future work, not implemented in this version.



Fig. 2.1 — image779.jpg: no EXIF data present; rotation unresolved.

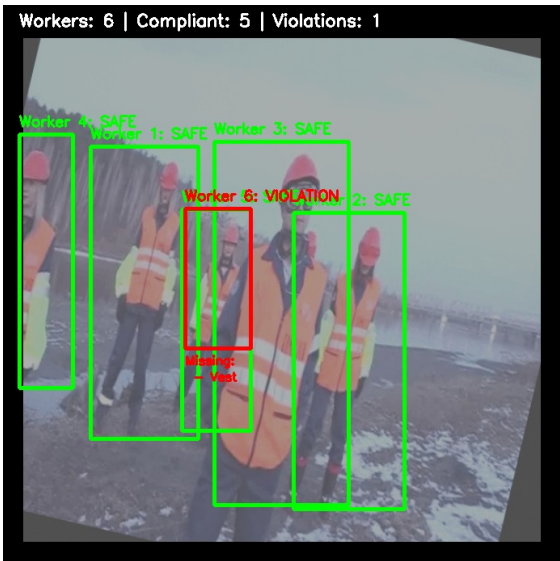


Fig. 2.2 — image462.jpg: additional rotated-feed instance, compounded by worker occlusion.

3. Top / Frame Boundary Truncation (Boundary Clipping)

STATUS: FIXED — Verified

OBSERVED BEHAVIOR	Workers positioned near the frame edge had PPE items (e.g. hard hats) partially truncated by the image border, reducing detection confidence and triggering false VIOLATION results.
MODEL PERCEPTION CHECK	Partial object truncation near boundaries measurably reduced detection confidence versus fully-visible equivalents.
ROOT-CAUSE MECHANICS	Objects near image boundaries receive incomplete visual context during inference, lowering model confidence below the effective detection threshold.
Production Fix	A 25-pixel padding border is applied to all sides of each image prior to inference (cv2.copyMakeBorder), giving boundary-adjacent objects additional surrounding context. Verified corrected on previously-affected evaluation image below.



Fig. 3.1 — image611.jpg: frame-edge cropping previously triggered false violation; resolved post-fix.

4. Overlapping / Occluded Workers (Multi-Person Spatial Ambiguity)

STATUS: DOCUMENTED — Fix Planned (v2)

OBSERVED BEHAVIOR	In scenes with multiple workers standing close together, PPE items belonging to one worker are sometimes not correctly associated with their body region, or occluded items are missed entirely, producing false VIOLATION results despite PPE being worn.
MODEL PERCEPTION CHECK	PPE items are frequently detected in the frame but fall outside the affected worker's calculated body-region boundaries due to bounding-box overlap between adjacent workers.
ROOT-CAUSE MECHANICS	The spatial region logic (head/torso/feet zones) is calculated independently per person box using fixed proportions. When person boxes overlap or a worker is partially occluded by another, PPE item positions can fall outside the intended worker's region, or be hidden entirely, causing misattribution.
Production Fix	Same underlying limitation as posture heuristics (Section 1). A pose-estimation-based approach with per-worker keypoint tracking would substantially reduce misattribution in crowded/occluded scenes. Not yet implemented.

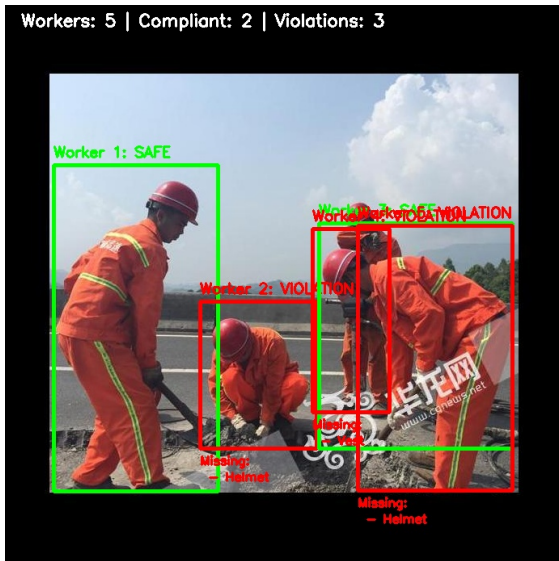


Fig. 4.1 — image116.jpg: overlapping workers causing PPE misattribution.

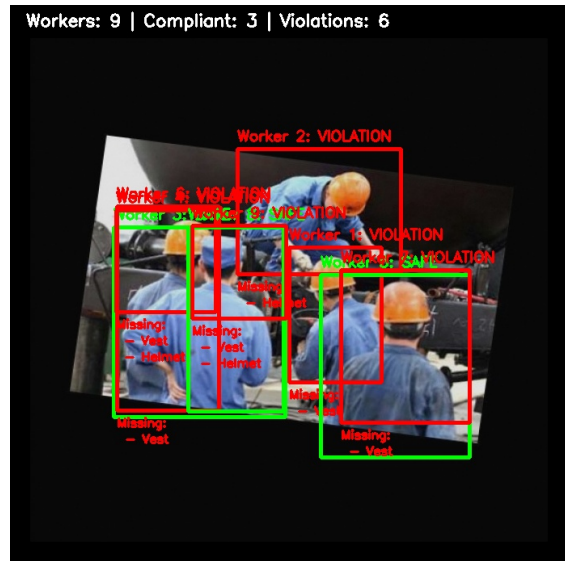


Fig. 4.2 — image450.jpg: dense multi-worker occlusion compounded by rotation.

5. Out-of-Domain Image Contamination (Data Quality / Distribution Shift)

STATUS: IDENTIFIED & QUANTIFIED — Data Curation Recommended

OBSERVED BEHAVIOR	High-confidence false-positive PPE detections (e.g. Helmet: 0.82, Vest: 0.80 confidence) occurred on images with no construction-site content at all — e.g. formal business settings, portraits, press-conference photography — which were present in the public source test dataset.
MODEL PERCEPTION CHECK	Model produced confident, well-formed bounding boxes on entirely out-of-domain content, with no built-in mechanism to reject inputs outside its training distribution.
ROOT-CAUSE MECHANICS	Manual review of the full 127-image test set identified 30 images (23.6%) as out-of-domain, mistakenly included in the public source dataset. This is a known, well-documented limitation of narrow-domain object detectors: without explicit out-of-distribution (OOD) rejection, models will produce confident predictions even on inputs entirely unlike their training data.
Production Fix	Two-part recommendation: (1) data curation — filter non-target-domain images from the source/test dataset prior to evaluation; (2) system-level — add a lightweight scene/domain classifier as a pre-filter stage to reject clearly out-of-domain inputs before running PPE detection. Neither implemented in this version; documented as a data-quality finding.



Fig. 5.1 — image1306.jpg: business-meeting photo, false PPE detections.



Fig. 5.2 — image1293.jpg: press-conference photo, false Person/violation flag.

6. Close-Up / Portrait-Style Framing (Non-Standard Body Proportions)

STATUS: DOCUMENTED

OBSERVED BEHAVIOR	Tightly-cropped, portrait-style images (e.g. stock photography with the worker's torso/head filling most of the frame) produce false VIOLATION results despite PPE being clearly visible and worn correctly.
MODEL PERCEPTION CHECK	PPE items (vest, helmet) are correctly detected with high confidence; the item simply falls outside the fixed-proportion body region calculated for that framing.
ROOT-CAUSE MECHANICS	The head/torso/feet region heuristic assumes standard full-body site-photo proportions. Tightly-cropped or portrait-oriented images have substantially different body-to-frame ratios, causing the fixed percentage boundaries to misalign with actual body regions — the same underlying category of failure as posture-based misalignment (Section 1), triggered by framing rather than pose.
Production Fix	Same production fix path as Sections 1 and 4: pose-based keypoint tracking would resolve this independently of framing or crop ratio. Alternatively, a framing/shot-type classifier could route non-standard crops to adjusted heuristics. Not yet implemented.



Fig. 6.1 — image718.jpg: close-up framing, vest clearly worn but flagged missing.



Fig. 6.2 — image169.jpg: additional close-up framing instance.

Comparative Analysis & Engineer's Summary

Scenario	YOLO Perception	Spatial Engine Result	Failure Domain	Status
Upright, isolated worker	Accurate (Conf > 0.85)	SAFE / Green Box	None	Production Ready
Bending / Crouching	Accurate	False Violation	Spatial Heuristic	Documented (v2 fix)
Rotated (no EXIF)	Accurate on rotated axis	False Violation	Data Availability	Partially Addressed
Frame Edge Crop	Low Confidence	False Violation	Boundary Clipping	Fixed
Overlapping Workers	Accurate, misattributed	False Violation	Multi-Person Ambiguity	Documented (v2 fix)
Out-of-Domain Image	False Positive (high conf.)	False Violation	Data Quality / OOD	Identified, Quantified
Close-up Framing	Accurate	False Violation	Non-Standard Proportions	Documented

Key Takeaway

One of six identified failure categories (frame-edge clipping) has been fully fixed and verified. One (out-of-domain contamination) has been quantified and root-caused as a data-quality issue rather than a modeling deficiency. The remaining four share a common underlying root cause — reliance on fixed-proportion spatial heuristics rather than pose-aware reasoning — and are documented with a consistent proposed fix (YOLOv8-Pose keypoint integration) scoped for a future version. This structured categorization reflects standard production ML practice: distinguishing model perception failures from downstream logic limitations, and data quality issues from either.